Introduction

This journal documents my 12-week placement as an IT Technician at a Further Education Campus. The placement aimed to bridge the gap between academic theory and professional practice. My core objectives included developing hardware diagnostic capabilities, mastering enterprise software deployment, and enhancing professional communication within a multi-campus IT infrastructure.

Week 1: Induction and Infrastructure Overview

The initial week focused on organizational integration. I was introduced to the IT support team and received an orientation of the campus infrastructure, including server rooms, localized switches, and the diverse range of end-user devices. A key component of this week was the induction into official IT policies, covering cybersecurity protocols, GDPR compliance, and Health and Safety standards for hardware handling.

I began practical tasks by assisting with the replacement of a Polar Team Pro docking station and syncing sensors for the sports department. I also shadowed senior technicians as they utilized the centralized ticketing system to prioritize daily tasks. By the end of the week, I performed my first hardware installations, including adding dualmonitor setups for staff workstations and replacing print heads on a specialized HP72 wide-format printer in room Engineering Lab A.

Week 2: Component-Level Hardware and Imaging

Week 2 involved more complex hardware interventions. I performed a full motherboard replacement on a Dell laptop, which required a systematic teardown and reassembly. This task emphasized the importance of electrostatic discharge (ESD) safety and meticulous cable management. Once the hardware was secured, I performed a POST (Power-On Self-Test) to ensure system integrity.

I also began working with the college's imaging system. I assisted in the deployment of standardized Windows 11 images to a batch of student laptops, ensuring all departmental software was pre-configured. Furthermore, I supported the "Idea Center" by troubleshooting connectivity for Oculus Quest 3 headsets, requiring me to navigate specialized VR software updates and Wi-Fi handshaking protocols.

Week 3: Service Desk Integration and Troubleshooting

During Week 3, I took a more active role in monitoring the Service Desk. This involved categorizing incoming tickets by urgency and department. I resolved several software-related issues, such as Microsoft 365 activation errors and forgotten password resets through Active Directory.

On the hardware side, I diagnosed a series of peripheral failures in the CAD labs. This involved testing USB ports and replacing faulty mechanical keyboards. I also conducted a "floor sweep" of the campus printers, replacing toner cartridges and cleaning rollers to prevent paper jams in high-traffic areas. These tasks highlighted the importance of preventative maintenance in reducing the total volume of support tickets.

Week 4: Advanced Hardware Repair and Professional Standards

In Week 4, I successfully completed a second motherboard replacement, demonstrating increased speed and confidence in my hardware diagnostics. I also handled a specialized ticket in computer lab A involving a malfunctioning plotter printer; this required a deep-clean of the print heads and a calibration of the paper feed sensors.

I spent time this week reviewing the college's data protection policies again to ensure that when I was "reimaging" staff laptops, I was strictly following data backup and wipe procedures. I ended the week by assisting in a bulk software push, manually ensuring that over 50 PCs in the main computer labs were powered on and ready to receive background security patches pushed from the central server.

Week 5: Systematic Auditing and Infrastructure Maintenance

This week marked the start of a major project: a campus-wide audit of laptop batteries and charging infrastructure. I systematically documented battery health, cycle counts, and physical condition for dozens of devices in two separate rooms. A critical finding was a misconfigured charging timer on a mobile trolley; I recalibrated the PDU (Power Distribution Unit) to a two-hour cycle to prevent overcharging.

This project emphasized the role of IT in long-term asset management. Beyond the audit, I provided audio-visual support in the Idea Center, calibrating a smart touchscreen for a guest speaker. This required me to explain technical setup steps to a non-technical user, which helped develop my professional communication skills.

Week 6: Classroom Support and Workstation Ergonomics

Week 6 focused on optimizing the learning environment. I assisted in the installation of five heavy-duty monitor stands in the Trading Room, which involved managing complex cable runs to maintain a clean and safe workspace. I also replaced faulty monitor power cables and installed a temporary smart touchscreen in the Innovations Center.

During my laptop audits on the first floor, I noted that devices were being mixed across three different trolleys, making asset tracking difficult. I proposed a more rigid labeling system to the senior team. I concluded the week by recalibrating several charging timers that had failed overnight, ensuring that 60+ laptops were ready for the Monday morning classes.

Week 7: Software Deployment and BIOS Security

In Week 7, I utilized the **UAS Master system** to deploy Microsoft .NET Framework 3.5 across several labs. This was essential for the compatibility of specialized engineering software. I also reimaged a Dell 7990 All-in One for a new staff member, which included updating the BIOS password to comply with the organization's high-security standards.

I also fixed a persistent charging problem in Laptop Trolley 16. After a physical inspection, I found that the issue was caused by a combination of a faulty timer and three damaged charging cables that had been frayed by the trolley door. This reinforced the need for frequent physical inspections of mobile IT equipment.

Week 8: Infrastructure Maintenance and Security Response

This week involved a fleet-wide maintenance sweep of Xerox printers in 11 different departments, ensuring black toner and waste cartridges were replaced before a busy exam period. More importantly, I assisted in a security incident response. After the central dashboard flagged "Trojan" alerts on eight devices in the Computer Lab D room, I isolated the machines and ran **Windows Defender Offline scans**. This provided a practical look at endpoint protection and the importance of network segmentation during a suspected breach.

Week 9: Software Imaging and Lab Readiness

With the campus on a reduced schedule for the holiday break, I focused on lab readiness. I worked in the workshop to reimage 20 student laptops using **PXE booting** to pull the latest "gold image" from the server. This ensured all security patches were current. I also set up a new staff workstation in Computer Lab B, which involved not just hardware installation but the mapping of specific departmental network drives and printer queues.

Week 10: Event Logistics and Hardware Failures

This week was dominated by the technical setup for the **Higher Education Open Day**. I configured Clever Touch displays for video loops and set up a bank of iPads for registration, troubleshooting Wi-Fi SSID authentication issues on the fly.

On the hardware side, I diagnosed a critical fan failure on an HP Z2 Tower G9 using BIOS error logs and initiated a warranty claim with the manufacturer. I also resolved a "No Boot Device" error on a staff PC by reconfiguring the UEFI boot priority, avoiding the need for a time-consuming reimage.

Week 11: Network Remediation and CLI Tools

Following a firmware update to the campus switches, 19 laptop trolleys lost their network "handshake." I spent the week visiting each location and using the **Command Line Interface (CLI)** to run `gpupdate /force` on over 300 laptops. This forced the devices to pull the updated Group Policy Objects (GPOs) and security certificates from the Domain Controller. This task required significant organization to track which labs were back online and which still faced connectivity hurdles.

Week 12: Final Troubleshooting and Vendor Liaison

In my final week, I resolved a DNS resolution failure in Laptop Trolley 7 by performing a DNS flush (`ipconfig /flushdns`) via the command line. I also performed a physical repair on a staff laptop by reseating a loose internal Wi-Fi antenna cable, which significantly improved signal strength.

I concluded my placement by collaborating with **Dell Support** to resolve BIOS password lockouts on four secure devices, providing Service Tags and proof of ownership to receive master override keys. Finally, I prepared a laser printer for an exam in room Classroom A, ensuring the toner was full and the drum was clean for highvolume printing. This week highlighted the transition from a supervised student to a technician capable of managing vendor relationships and complex network troubleshooting.